Factorization Systems for Dialectica Categories

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Abstract

The Dialectica construction is a family of constructions: the original one, Dial($\mathcal{C}$), whose backward maps see the witness, and the Girard one, Gir($\mathcal{C}$), whose backward maps do not. We describe the orthogonal factorization systems that both members of the family inherit from a factorization system on the base category $\mathcal{C}$. Forgetting the relations sends Gir($\mathcal{C}$) faithfully to $\mathcal{C} \times \mathcal{C}^{op}$, and a factorization system on $\mathcal{C}$ lifts in two ways, according to whether the middle object of a factorization carries the image relation or the largest possible relation. Over sets the two lifts are the systems of regular epimorphisms and monomorphisms, and of epimorphisms and regular monomorphisms; the system of all epimorphisms and monomorphisms does not exist, since Gir(**Sets**) is not balanced. Forgetting the relations of Dial($\mathcal{C}$) lands instead in a category whose backward maps depend on witnesses, and there only the witness side of a morphism can be factored. We describe the resulting systems and compute the epimorphisms and monomorphisms of Dial(**Sets**).

1 Introduction

Several statements about dialectica categories only make sense relative to a choice of which maps count as "surjections" and which as "injections". The clearest examples come from set theory. Samuel da Silva showed that the method of morphisms for cardinal invariants, read in full generality, is equivalent to the Partition Principle, while the existence of a weakly initial object in the categories PV_X is equivalent to the Axiom of Choice [3, 4]. Categorical versions of choice and of the partition principle are usually phrased as "every epimorphism splits" or "every epimorphism has a monomorphism going back". Stated for all epimorphisms and monomorphisms, these are intrinsic to the category. Their reformulations in terms of a chosen class of surjections and a chosen class of injections depend on a factorization system, and in a category that is not balanced, in the sense recalled in Section 2, the epimorphisms and the monomorphisms do not form one.

The dialectica categories come from the categorical reading of Gödel's Dialectica interpretation [7, 5]. Gödel's interpretation turns a formula into a statement about witnesses and counterexamples, and the categorical version keeps exactly this data. An object is a triple (U, X, α), with U an object of witnesses, X an object of counterexamples and α a relation between them, saying which witnesses withstand which counterexamples. A morphism sends witnesses forward and counterexamples backward, and must preserve the relation: if a witness u withstands the counterexample sent back from y, then the image of u withstands y. How much the backward map may see distinguishes the two members of the Dialectica family studied in the thesis [5]. In the original construction Dial($\mathcal{C}$) the backward map may use the witness, as in Gödel's interpretation of implication; in the Girard construction Gir($\mathcal{C}$) it ignores the witness. Both are models of linear logic, Dial($\mathcal{C}$) of its intuitionistic fragment and Gir($\mathcal{C}$) of the classical one. A close relative of Gir(**Sets**), Blass's category of problems and solutions, is used in set theory, where morphisms between such triples express inequalities between cardinal invariants [1, 4].

A factorization system on a category splits every morphism into two parts of prescribed kinds, the way a function between sets factors as a surjection onto its image followed by the inclusion of the image. The two kinds are given by two classes of maps, which in an orthogonal factorization system determine each other, and the factorization of each map is unique up to a unique isomorphism; the precise definition is recalled in Section 2. Besides surjections and injections in **Sets**, familiar examples are the regular epimorphisms and monomorphisms of any regular category and, in topological spaces, both the quotient maps with the injective continuous maps and the surjective continuous maps with the embeddings. One category can therefore carry several factorization systems, and the statements of choice and partition principles above change with the system chosen.

This note records the factorization systems that both members of the Dialectica family inherit from one on $\mathcal{C}$. The Girard construction behaves well because forgetting its relations gives a faithful functor to $\mathcal{C} \times \mathcal{C}^{op}$, along which relations can be pushed forward and pulled back: both witnesses and counterexamples can be factored, and the relation is then placed on the middle object in one of two canonical ways. Over sets this reproduces, with the relation in the role of the topology, the two systems of topological spaces just mentioned. The original construction only allows the witness side to be factored, because the counterexample component of a morphism depends on the witness.

2 Dialectica categories and factorization systems

We recall the two members of the Dialectica family, the categories they sit over, and the definition of a factorization system, since the factorization systems of this note are built on those base categories and then lifted. Let $\mathcal{C}$ be a category with finite limits. A dialectica object is a triple (U, X, α) with α a subobject of $U \times X$; we call U the witnesses and X the counterexamples. Both constructions below are from the thesis [5]. We read the relations constructively throughout: over **Sets** the arguments use only that **Sets** is a topos, and never that a relation equals its double negation.

A morphism $(f, F)\colon (U, X, \alpha) \to (V, Y, \beta)$ of $\mathrm{Gir}(\mathcal{C})$ consists of $f\colon U \to V$ and $F\colon Y \to X$ such that

$$(U \times F)^*\alpha \;\leq\; (f \times Y)^*\beta \quad \text{in } \mathrm{Sub}(U \times Y).$$

In **Sets** this says that $u\,\alpha\,F(y)$ implies $f(u)\,\beta\,y$. Composition is $(g, G) \circ (f, F) = (gf,\, FG)$.

A morphism $(f, F)\colon (U, X, \alpha) \to (V, Y, \beta)$ of $\mathrm{Dial}(\mathcal{C})$ consists of $f\colon U \to V$ and $F\colon U \times Y \to X$ such that

$$\langle \pi_1, F\rangle^*\alpha \;\leq\; (f \times Y)^*\beta \quad \text{in } \mathrm{Sub}(U \times Y),$$

that is, $u\,\alpha\,F(u, y)$ implies $f(u)\,\beta\,y$. Composition is $(g, G) \circ (f, F) = (gf,\, (u, z) \mapsto F(u, G(fu, z)))$.

Forgetting the relations gives faithful functors

$$p\colon \mathrm{Gir}(\mathcal{C}) \to \mathcal{C} \times \mathcal{C}^{op}, \qquad q\colon \mathrm{Dial}(\mathcal{C}) \to \mathrm{Dial}_{\mathrm{base}}(\mathcal{C}),$$

where $\mathrm{Dial}_{\mathrm{base}}(\mathcal{C})$ has the same objects as $\mathcal{C} \times \mathcal{C}^{op}$, pairs (U, X), and as morphisms pairs (f, F) with $f\colon U \to V$ and $F\colon U \times Y \to X$, composed as in $\mathrm{Dial}(\mathcal{C})$. It is $\mathcal{C} \times \mathcal{C}^{op}$ with the backward component taken in the coKleisli category of the comonad $U \times (-)$, so that the backward map may use the witness, and $\mathcal{C} \times \mathcal{C}^{op}$ maps into it by letting the backward component ignore the witness; this functor is faithful on morphisms out of objects with inhabited witnesses, as discussed in Section 6. In passing, for $\mathcal{C} = \mathbf{Sets}$ the category $\mathrm{Dial}_{\mathrm{base}}(\mathbf{Sets})$ is the category of monomials Uy^X among polynomial functors [10]. In both cases the fibre over (U, X) is the poset $\mathrm{Sub}(U \times X)$, and a vertical map $(\mathrm{id}, \mathrm{id})\colon (U, X, \alpha) \to (U, X, \beta)$ exists exactly when $\alpha \leq \beta$.

Definition 2.1. An *orthogonal factorization system* on a category $\mathbb{B}$ is a pair $(\mathcal{L}, \mathcal{R})$ of classes of morphisms of $\mathbb{B}$ such that

1. both classes contain the isomorphisms and are closed under composition with them;

2. every morphism h factors as $h = r \circ l$ with $l \in \mathcal{L}$ and $r \in \mathcal{R}$;

3. for every commutative square $r \circ u = v \circ l$ with $l \in \mathcal{L}$ and $r \in \mathcal{R}$ there is exactly one morphism d with $d \circ l = u$ and $r \circ d = v$.

The third condition says that every map in $\mathcal{L}$ is orthogonal to every map in $\mathcal{R}$. It makes factorizations unique up to a unique isomorphism, and it implies that both classes are closed under composition and that each class determines the other [6]. Over **Sets** the surjections and the injections form an orthogonal factorization system; in a category with such a system we think of the maps in $\mathcal{L}$ as surjections and of those in $\mathcal{R}$ as injections.

A category is balanced when every morphism that is both a monomorphism and an epimorphism is an isomorphism. Sets and every topos are balanced, while categories of sets with structure often are not: in topological spaces, the identity from a discrete two-point space onto Sierpiński space is a monomorphism and an epimorphism but not a homeomorphism. In any orthogonal factorization system the maps belonging to both classes are exactly the isomorphisms, so in a category that is not balanced the epimorphisms and the monomorphisms cannot form a factorization system.

3 The Girard version of Dialectica

The Girard construction is the easier case, because forgetting the relations gives a faithful functor to $\mathcal{C} \times \mathcal{C}^{op}$ along which relations can be pushed forward and pulled back. The relational part of a factorization is handled by the following lifts, which use images and universal quantification along maps in $\mathcal{C}$.

Lemma 3.1. *Let $(f, F)\colon (U, X) \to (V, Y)$ be a map of $\mathcal{C} \times \mathcal{C}^{op}$.*

1. *If $\mathcal{C}$ is regular, the relation $\beta_* = \exists_{f \times Y}(U \times F)^* \alpha$ makes $(f, F)\colon (U, X, \alpha) \to (V, Y, \beta_*)$ a p-cocartesian morphism.*

2. *If $\mathcal{C}$ is a Heyting category, the relation $\alpha^* = \forall_{U \times F}(f \times Y)^* \beta$ makes $(f, F)\colon (U, X, \alpha^*) \to (V, Y, \beta)$ a p-cartesian morphism.*

Hence p is an opfibration in the first case and a bifibration in the second.

Proof. The relation β_* is the least β making (f, F) a morphism, by the adjunction $\exists_{f \times Y} \dashv (f \times Y)^*$. For cocartesianness, take $(g, G)\colon (V, Y) \to (W, Z)$ and suppose $(gf, FG)\colon (U, X, \alpha) \to (W, Z, \gamma)$ is a morphism. The square formed by $f \times Z$, $U \times G$, $V \times G$ and $f \times Y$ is a pullback, so Beck–Chevalley gives

$$(V \times G)^* \beta_* = \exists_{f \times Z}(U \times G)^* (U \times F)^* \alpha = \exists_{f \times Z}(U \times FG)^* \alpha.$$

This is below $(g \times Z)^* \gamma$ exactly when $(U \times FG)^* \alpha \leq (gf \times Z)^* \gamma$, which is the hypothesis. The cartesian case is dual, using $(U \times F)^* \dashv \forall_{U \times F}$ and Beck–Chevalley for $\forall$. □

Since the fibres are posets, these lifts are unique and can be recognised from the relations alone. A morphism $(f, F)\colon (U, X, \alpha) \to (V, Y, \beta)$ is cocartesian exactly when $\beta = \beta_*$, and cartesian exactly when $\alpha = \alpha^*$: in each case the morphism and the lift lie over the same base map and differ by a vertical isomorphism, and vertical isomorphisms between objects of a poset fibre are identities.

The same lifts exist when relations take values in a complete lineale L in the sense of Hyland and de Paiva [9], or indeed in any complete lattice. There a morphism satisfies $\alpha(u, Fy) \leq \beta(fu, y)$, and the least β and the largest α satisfying this inequality are

$$\beta_*(v, y) = \bigvee\{\alpha(u, Fy) : fu = v\}, \qquad \alpha^*(u, x) = \bigwedge\{\beta(fu, y) : Fy = x\},$$

with the empty join equal to the bottom and the empty meet equal to the top. Only completeness of the lattice is used.

A factorization system on the base lifts along an opfibration, or dually along a fibration. We record the standard argument.

Lemma 3.2. *Let $p\colon \mathbb{E} \to \mathbb{B}$ be an opfibration and $(\mathcal{L}, \mathcal{R})$ an orthogonal factorization system on $\mathbb{B}$. Let $\mathcal{L}'$ be the class of cocartesian morphisms h with $p(h) \in \mathcal{L}$, and $\mathcal{R}'$ the class of all h with $p(h) \in \mathcal{R}$. Then $(\mathcal{L}', \mathcal{R}')$ is an orthogonal factorization system on $\mathbb{E}$. Dually, if p is a fibration, the class of all h over $\mathcal{L}$ and the class of cartesian h over $\mathcal{R}$ form one.*

Proof. Both classes contain the isomorphisms and are closed under composition with them. Given $h\colon A \to B$, factor $p(h) = r \circ l$ in $\mathbb{B}$ and take a cocartesian lift $\hat{l}\colon A \to l_!A$. Cocartesianness gives a unique $\hat{r}$ over r with $h = \hat{r}\hat{l}$. For orthogonality, let $\hat{l}\colon A \to B$ be in $\mathcal{L}'$, $\hat{r}\colon C \to D$ in $\mathcal{R}'$, and $\hat{r}u = v\hat{l}$. The base square has a unique diagonal d, and cocartesianness of $\hat{l}$ gives a unique δ over d with $\delta\hat{l} = u$. The maps $\hat{r}\delta$ and v lie over the same base map and agree after precomposition with $\hat{l}$, so they are equal. Any other diagonal lies over d by uniqueness in the base and then equals δ. The usual characterization of orthogonal factorization systems [6] finishes the proof. □

A factorization system $(\mathcal{E}, \mathcal{M})$ on $\mathcal{C}$ gives the system $(\mathcal{E} \times \mathcal{M}^{op}, \mathcal{M} \times \mathcal{E}^{op})$ on $\mathcal{C} \times \mathcal{C}^{op}$. Explicitly, (f, F) is in the left class when $f \in \mathcal{E}$ and $F \in \mathcal{M}$, and in the right class when $f \in \mathcal{M}$ and $F \in \mathcal{E}$: the roles swap on the counterexample side because F runs backwards. Combining with the two lemmas gives the main statement.

Theorem 3.3. *Let $(\mathcal{E}, \mathcal{M})$ be an orthogonal factorization system on $\mathcal{C}$.*

(I) If $\mathcal{C}$ is regular, $\mathrm{Gir}(\mathcal{C})$ *has the factorization system whose left class consists of the p-cocartesian morphisms (f, F) with $f \in \mathcal{E}$ and $F \in \mathcal{M}$, and whose right class consists of all morphisms with $f \in \mathcal{M}$ and $F \in \mathcal{E}$.*

(II) If $\mathcal{C}$ is a Heyting category, $\mathrm{Gir}(\mathcal{C})$ *has the factorization system whose left class consists of all morphisms with $f \in \mathcal{E}$ and $F \in \mathcal{M}$, and whose right class consists of the p-cartesian morphisms with $f \in \mathcal{M}$ and $F \in \mathcal{E}$.*

Concretely, factor $f = m \circ e$ through W and $F = m' \circ e'$ through Z. In system (I) the middle object is (W, Z, γ_*) with γ_* the image of α, and in system (II) it is (W, Z, γ^*) with γ^* the largest relation for which the second half is a morphism. When $\mathcal{C}$ is a Heyting category, every morphism factors in three steps: a cocartesian map over $\mathcal{E} \times \mathcal{M}^{op}$, a vertical map $\gamma_* \leq \gamma^*$, and a cartesian map over $\mathcal{M} \times \mathcal{E}^{op}$. The two systems differ only in which side absorbs the vertical step, and any relation between γ_* and γ^* gives a factorization, though not an orthogonal one.

Proposition 3.4. *Suppose $\mathcal{C}$ is a Heyting category. The left class of (I) is contained in the left class of (II), and the right class of (II) is contained in the right class of (I). These inclusions are equalities if and only if every fibre* $\mathrm{Sub}(U \times X)$ *has a single element.*

Proof. Cocartesian maps over $\mathcal{E} \times \mathcal{M}^{op}$ are maps over $\mathcal{E} \times \mathcal{M}^{op}$, and cartesian maps over $\mathcal{M} \times \mathcal{E}^{op}$ are maps over $\mathcal{M} \times \mathcal{E}^{op}$, which gives the inclusions. If every fibre is a singleton, every morphism is cartesian and cocartesian, so the classes agree. Conversely, let $\alpha \leq \beta$ in $\mathrm{Sub}(U \times X)$. The vertical map $(\mathrm{id}, \mathrm{id})\colon (U, X, \alpha) \to (U, X, \beta)$ lies over an isomorphism, hence in the left class of (II). If this equals the left class of (I), the map is cocartesian over the identity, which forces $\beta = \exists_{\mathrm{id}} \alpha = \alpha$. Taking β to be the top subobject shows that every fibre is a singleton. □

In the usual order on factorization systems, by inclusion of left classes, system (I) therefore lies below system (II). Over **Sets** this is the familiar comparison of (regular epimorphisms, monomorphisms) with (epimorphisms, regular monomorphisms), as in topological spaces.

The two systems together determine most of the structure that the relations add to $\mathrm{Gir}(\mathcal{C})$. Since $\mathcal{E} \cap \mathcal{M}$ consists of the isomorphisms of $\mathcal{C}$, the intersection $\mathcal{L}_{II} \cap \mathcal{R}_I$ consists exactly of the morphisms lying over isomorphisms of $\mathcal{C} \times \mathcal{C}^{op}$, that is, of the vertical morphisms up to a change of the underlying objects, and these recover the order on each fibre. The other intersection, $\mathcal{L}_I \cap \mathcal{R}_{II}$, consists only of the isomorphisms, and the difference between the two intersections is the vertical step $\gamma_* \leq \gamma^*$ described above. The base can be recovered as well. The functor p has fully faithful adjoints on both sides, the left adjoint giving a pair (U, X) the least relation and the right adjoint giving it the full relation, and by a result of Gabriel and Zisman [8] a functor with a fully faithful adjoint is the localization at the morphisms it inverts. The morphisms that p inverts are those lying over isomorphisms, so $\mathcal{C} \times \mathcal{C}^{op}$ is equivalent to $\mathrm{Gir}(\mathcal{C})$ with the morphisms of $\mathcal{L}_{II} \cap \mathcal{R}_I$ inverted. Finally, $\mathcal{L}_I$ picks out the cocartesian morphisms within $\mathcal{L}_{II}$, and $\mathcal{R}_{II}$ the cartesian morphisms within $\mathcal{R}_I$, so the pair of systems also sees the two lifts of Lemma 3.1.

4 The Girard version over sets

Over sets the two systems of Theorem 3.3, starting from the epi–mono system of **Sets**, are familiar ones. To identify them we compute the epimorphisms, monomorphisms and their regular versions in $\mathrm{Gir}(\mathbf{Sets})$, using only constructive arguments as announced in Section 2.

Proposition 4.1. *A morphism (f, F) of $\mathrm{Gir}(\mathbf{Sets})$ is*

1. *an epimorphism if and only if f is surjective and F is injective;*
2. *a monomorphism if and only if f is injective and F is surjective;*
3. *a regular epimorphism if and only if it is an epimorphism and p-cocartesian;*
4. *a regular monomorphism if and only if it is a monomorphism and p-cartesian.*

Proof. Sufficiency in (1) and (2) is immediate from the composition law, since p is faithful. For necessity in (2), maps out of $(1, 1, \emptyset)$ satisfy the morphism condition vacuously and pick out arbitrary witnesses, which forces f to be injective. Maps out of $(\emptyset, \Omega, \emptyset)$, where Ω is the set of truth values, have arbitrary counterexample components $X \to \Omega$. The map sending every x to $\top$ and the map sending x to the truth value of $\exists y\, F(y) = x$ agree after composition with F, so they are equal, and F is surjective. For necessity in (1), maps into $(\Omega, \emptyset, \emptyset)$, whose condition is vacuous, force f to be surjective by the same argument with the maps $v \mapsto \top$ and $v \mapsto (\exists u\, f(u) = v)$, and maps into $(1, 1, \top)$, whose condition always holds, force F to be injective.

For (3), note first that p has a left adjoint, sending (U, X) to $(U, X, \emptyset)$, and a right adjoint, sending (V, Y) to $(V, Y, \top)$: every map of the base is a morphism out of an object with the empty relation and into an object with the full relation. Hence p preserves limits and colimits.

Suppose $(f,F)\colon (U,X,\alpha)\to(V,Y,\beta)$ is the coequalizer of a parallel pair $(g_1,G_1),(g_2,G_2)\colon (T,S,\gamma)\to(U,X,\alpha)$. Its image under p is a coequalizer in $\mathbf{Sets}\times\mathbf{Sets}^{op}$, so f is a coequalizer in $\mathbf{Sets}$ and F an equalizer in $\mathbf{Sets}$; in particular f is surjective and F is injective. Let β_* be the relation given by Lemma 3.1, so that $v\,\beta_*\,y$ holds exactly when $u\,\alpha\,F(y)$ for some u with $f(u)=v$. Since (f,F) is a morphism into (V,Y,β), we have $\beta_*\le\beta$. The cocartesian lift $(f,F)\colon(U,X,\alpha)\to(V,Y,\beta_*)$ coequalizes the same pair, because composites are computed in the base. The universal property gives a morphism $(g,G)\colon(V,Y,\beta)\to(V,Y,\beta_*)$ with $gf=f$ and $FG=F$. Since f is surjective and F is injective, g and G are identities. That $(\mathrm{id},\mathrm{id})$ is a morphism from (V,Y,β) to (V,Y,β_*) says $\beta\le\beta_*$. Hence $\beta=\beta_*$, and (f,F) is cocartesian.

Conversely, let $(f,F)\colon(U,X,\alpha)\to(V,Y,\beta)$ be cocartesian, with f surjective and F injective. Let $k_1,k_2\colon K\to U$ be the kernel pair of f and $c_1,c_2\colon X\to P$ the cokernel pair of F. Each $(k_i,c_i)\colon(K,P,\emptyset)\to(U,X,\alpha)$ is a morphism, since its condition is vacuous, and $(f,F)\circ(k_i,c_i)=(fk_i,\,c_iF)$ does not depend on i. We check that (f,F) is their coequalizer. Let $(h,H)\colon(U,X,\alpha)\to(W,Z,\delta)$ satisfy $hk_1=hk_2$ and $c_1H=c_2H$. In $\mathbf{Sets}$ a surjection is the coequalizer of its kernel pair, so there is a unique $g\colon V\to W$ with $gf=h$. An injection is the equalizer of its cokernel pair, also when its domain is empty, so there is a unique $G\colon Z\to Y$ with $FG=H$. Uniqueness of (g,G) in $\mathrm{Gir}(\mathbf{Sets})$ follows because p is faithful. It remains to show that $(g,G)\colon(V,Y,\beta)\to(W,Z,\delta)$ is a morphism. Suppose $v\,\beta\,G(z)$. Since $\beta=\beta_*$, there is u with $f(u)=v$ and $u\,\alpha\,F(G(z))$, that is, $u\,\alpha\,H(z)$. As (h,H) is a morphism, $h(u)\,\delta\,z$, and $h(u)=g(f(u))=g(v)$. Hence $g(v)\,\delta\,z$.

For (4), suppose first that $(f,F)\colon(U,X,\alpha)\to(V,Y,\beta)$ is the equalizer of a parallel pair $(h_1,H_1),(h_2,H_2)\colon(V,Y,\beta)\to(W,Z,\delta)$. Its image under p is an equalizer in $\mathbf{Sets}\times\mathbf{Sets}^{op}$, so f is an equalizer and F a coequalizer in $\mathbf{Sets}$; in particular f is injective and F is surjective. Let α^* be the relation given by Lemma 3.1, so that $u\,\alpha^*\,x$ holds exactly when $f(u)\,\beta\,y$ for every y with $F(y)=x$. Since (f,F) is a morphism from (U,X,α), we have $\alpha\le\alpha^*$. The cartesian lift $(f,F)\colon(U,X,\alpha^*)\to(V,Y,\beta)$ equalizes the same pair, so there is a morphism $(g,G)\colon(U,X,\alpha^*)\to(U,X,\alpha)$ with $fg=f$ and $GF=F$. Since f is injective and F is surjective, g and G are identities, and that $(\mathrm{id},\mathrm{id})$ is a morphism from (U,X,α^*) to (U,X,α) says $\alpha^*\le\alpha$. Hence $\alpha=\alpha^*$, and (f,F) is cartesian.

Conversely, let $(f,F)\colon(U,X,\alpha)\to(V,Y,\beta)$ be cartesian, with f injective and F surjective. Let $c_1,c_2\colon V\to Q$ be the cokernel pair of f and $k_1,k_2\colon L\to Y$ the kernel pair of F. Each $(c_i,k_i)\colon(V,Y,\beta)\to(Q,L,\top)$ is a morphism, since its condition always holds, and $(c_i,k_i)\circ(f,F)=(c_if,\,Fk_i)$ does not depend on i. Let $(h,H)\colon(T,S,\gamma)\to(V,Y,\beta)$ satisfy $c_1h=c_2h$ and $Hk_1=Hk_2$. Since f is the equalizer of its cokernel pair, there is a unique $g\colon T\to U$ with $fg=h$, and since F is the coequalizer of its kernel pair, there is a unique $G\colon X\to S$ with $GF=H$. It remains to show that $(g,G)\colon(T,S,\gamma)\to(U,X,\alpha)$ is a morphism. Suppose $t\,\gamma\,G(x)$. Since $\alpha=\alpha^*$, we must show $f(g(t))\,\beta\,y$ for every y with $F(y)=x$. For such y we have $G(x)=G(F(y))=H(y)$, so $t\,\gamma\,H(y)$, and since (h,H) is a morphism, $h(t)\,\beta\,y$, where $h(t)=f(g(t))$. □

Part (4) is proved directly, and not deduced from (3) by duality, because the duality is not an involution when relations are read constructively. Sending (U,X,α) to $(X,U,\alpha^\perp)$, where $x\,\alpha^\perp\,u$ is the negation of $u\,\alpha\,x$, and (f,F) to (F,f), defines a functor $N\colon\mathrm{Gir}(\mathbf{Sets})^{op}\to\mathrm{Gir}(\mathbf{Sets})$, since the morphism condition for (F,f) is the contrapositive of the one for (f,F). Up to isomorphism $N(A)$ is the linear negation $A\multimap\perp$, with $\perp=(1,1,\emptyset)$. Applying N twice gives $(U,X,\neg\neg\alpha)$, and since $\alpha\le\neg\neg\alpha$ the identities form a morphism from A to $N(N(A))$, which is an isomorphism exactly when α is $\neg\neg$-closed. Because $\neg\exists$ is $\forall\neg$, the functor N sends cocartesian morphisms to cartesian ones, while the converse holds only when the relations involved are $\neg\neg$-closed.

Corollary 4.2. *Starting from the epi–mono system of* **Sets**, *system (I) on* $\mathrm{Gir}(\mathbf{Sets})$ *is (regular epimorphisms, monomorphisms) and system (II) is (epimorphisms, regular monomorphisms).*

Vertical maps $(\mathrm{id},\mathrm{id})\colon (U,X,\alpha)\to(U,X,\beta)$ *with* $\alpha<\beta$ *are epimorphisms and monomorphisms but not isomorphisms, so* $\mathrm{Gir}(\mathbf{Sets})$ *is not balanced and has no (epimorphism, monomorphism) factorization system.*

The analogy with topological spaces is close, with the relation playing the role of the topology. In fact $p\colon \mathrm{Gir}(\mathbf{Sets})\to \mathbf{Sets}\times\mathbf{Sets}^{op}$ is a topological functor in the sense of Adámek, Herrlich and Strecker [2]: fibres are small complete lattices, and the initial lift of a source is the meet of the cartesian lifts. In particular $\mathrm{Gir}(\mathbf{Sets})$ is complete and cocomplete, with limits and colimits computed in $\mathbf{Sets}\times\mathbf{Sets}^{op}$ and relations given by initial and final lifts.

Regular epimorphisms of $\mathrm{Gir}(\mathbf{Sets})$ need not split, even though **Sets** satisfies the Axiom of Choice. The obstruction comes from the relations.

Example 4.3. Let $A=(\{a,b\},\{x_1,x_2\},\alpha)$ with $\alpha=\{(a,x_1),(b,x_2)\}$, and $B=(1,1,\top)$. The map $h=(!,F)\colon A\to B$ with $F(*)=x_1$ is a morphism, it is cocartesian because $a\,\alpha\,x_1$, its witness component is surjective, and its counterexample component is injective; so h is a regular epimorphism. A morphism $(s,G)\colon B\to A$ would need an element $s(*)$ related by α to both x_1 and x_2, and there is none. Hence there are no morphisms from B to A at all, so h has neither a section nor a monomorphism going back.

5 The Gödel version of Dialectica

Forgetting the relations of $\mathrm{Dial}(\mathcal{C})$ gives the faithful functor q to $\mathrm{Dial}_{\mathrm{base}}(\mathcal{C})$, and here only the witness side factors well. For a morphism $(f,F)\colon (U,X)\to(V,Y)$ of $\mathrm{Dial}_{\mathrm{base}}(\mathcal{C})$ we say that F is fibrewise invertible when $\langle\pi_1,F\rangle\colon U\times Y\to U\times X$ is an isomorphism. Over **Sets** this says that every $F(u,-)\colon Y\to X$ is a bijection. We write morphisms of $\mathcal{C}$ with variables, so that $F(u,y)$ stands for the composite of F with a generalized element (u,y) of $U\times Y$; every equation below is an equation between morphisms of $\mathcal{C}$, and only finite products are used.

Lemma 5.1. *Let* $(\mathcal{E},\mathcal{M})$ *be an orthogonal factorization system on* $\mathcal{C}$*. In* $\mathrm{Dial}_{\mathrm{base}}(\mathcal{C})$ *the classes*

$$\mathcal{L}_{\mathcal{E}}=\{(f,F): f\in\mathcal{E}\},\qquad \mathcal{R}_{\mathcal{M}}=\{(f,F): f\in\mathcal{M},\ F \textit{ fibrewise invertible}\}$$

form an orthogonal factorization system.

Proof. For the factorization, write $f=m\circ e$ through W. Then $(f,F)=(m,\pi_2)\circ(e,F)$, with middle object (W,Y) and $\pi_2\colon W\times Y\to Y$ the projection: the composite has witness component $me=f$ and counterexample component $(u,y)\mapsto F(u,\pi_2(e(u),y))=F(u,y)$. The first factor lies in $\mathcal{L}_{\mathcal{E}}$, and the second lies in $\mathcal{R}_{\mathcal{M}}$ because $\langle\pi_1,\pi_2\rangle$ is the identity. Both classes contain the isomorphisms and are closed under composition, since $\langle\pi_1,-\rangle$ of a composite is a composite of maps of the form $\langle\pi_1,-\rangle$ and their pullbacks, and their intersection consists of the isomorphisms.

For orthogonality, consider $l=(e,L)\colon (A_0,A_1)\to(B_0,B_1)$ in $\mathcal{L}_{\mathcal{E}}$, $r=(m,R)\colon (C_0,C_1)\to(D_0,D_1)$ in $\mathcal{R}_{\mathcal{M}}$, and $u=(u_0,U)$, $v=(v_0,V)$ with $r\circ u=v\circ l$. By the composition law this says that $mu_0=v_0e$ and that

$$U\big(a,R(u_0a,\delta)\big)=L\big(a,V(ea,\delta)\big).$$

A diagonal $d=(d_0,D)\colon (B_0,B_1)\to(C_0,C_1)$ must satisfy $md_0=v_0$ and $d_0e=u_0$ on witnesses, and since $e\in\mathcal{E}$ and $m\in\mathcal{M}$ there is exactly one such d_0. For any $t\colon T\to C_0$, the map $\langle\pi_1,R(t\times D_1)\rangle\colon T\times D_1\to T\times C_1$ is the pullback of the isomorphism $\langle\pi_1,R\rangle$ along $t\times C_1$, hence itself an isomorphism. Write ρ for this isomorphism when $t=d_0$ and σ when $t=u_0$. On counterexamples, the equation $r\circ d=v$ says that $D\circ\rho=V$, so it has exactly one solution, $D=V\circ\rho^{-1}$. It remains to

check $d \circ l = u$. The witness components agree because $d_0 e = u_0$. The counterexample component of $d \circ l$ is $L \circ \langle \pi_1, D(e \times C_1) \rangle$, and since σ is an isomorphism it suffices to compare both sides after composing with σ. Using $d_0 e = u_0$ and $D \circ \rho = V$,

$$D\big(ea, R(u_0 a, \delta)\big) = D\big(ea, R(d_0 ea, \delta)\big) = V(ea, \delta),$$

so $L \circ \langle \pi_1, D(e \times C_1) \rangle \circ \sigma$ sends (a, δ) to $L(a, V(ea, \delta))$, while $U \circ \sigma$ sends it to $U(a, R(u_0 a, \delta))$. These agree by the commutativity of the square. The argument uses inverses of isomorphisms and nothing about double negations, so it is valid constructively. □

The relational part works as for the Girard version, with $\langle \pi_1, F \rangle$ in place of $U \times F$. The only new point is that the pullback squares needed for Beck–Chevalley now involve backward maps that see the witnesses.

Lemma 5.2. *Let $(f, F) \colon (U, X) \to (V, Y)$ be a morphism of $\mathrm{Dial}_{\mathrm{base}}(\mathcal{C})$.*

1. *If $\mathcal{C}$ is regular, the relation $\beta_* = \exists_{f \times Y} \langle \pi_1, F \rangle^* \alpha$ makes $(f, F) \colon (U, X, \alpha) \to (V, Y, \beta_*)$ a q-cocartesian morphism.*

2. *If $\mathcal{C}$ is a Heyting category, the relation $\alpha^* = \forall_{\langle \pi_1, F \rangle} (f \times Y)^* \beta$ makes $(f, F) \colon (U, X, \alpha^*) \to (V, Y, \beta)$ a q-cartesian morphism.*

Hence q is an opfibration in the first case and a bifibration in the second.

Proof. By the adjunctions $\exists_{f \times Y} \dashv (f \times Y)^*$ and $\langle \pi_1, F \rangle^* \dashv \forall_{\langle \pi_1, F \rangle}$, the relation β_* is the least β, and α^* the largest α, making (f, F) a morphism.

For cocartesianness, take $(g, G) \colon (V, Y) \to (W, Z)$ and suppose the composite (gf, H), with $H(u, z) = F(u, G(fu, z))$, is a morphism from (U, X, α) to (W, Z, γ). The square with sides $f \times Z \colon U \times Z \to V \times Z$, $\langle \pi_1, G(f \times Z) \rangle \colon U \times Z \to U \times Y$, $\langle \pi_1, G \rangle \colon V \times Z \to V \times Y$ and $f \times Y \colon U \times Y \to V \times Y$ is a pullback: a pair of generalized elements (v, z) and (u, y) with $(v, G(v, z)) = (fu, y)$ is determined by (u, z). Beck–Chevalley for $\exists$ in a regular category gives

$$\langle \pi_1, G \rangle^* \beta_* = \exists_{f \times Z} \langle \pi_1, G(f \times Z) \rangle^* \langle \pi_1, F \rangle^* \alpha = \exists_{f \times Z} \langle \pi_1, H \rangle^* \alpha,$$

since $\langle \pi_1, F \rangle \circ \langle \pi_1, G(f \times Z) \rangle = \langle \pi_1, H \rangle$. This is below $(g \times Z)^* \gamma$ exactly when $\langle \pi_1, H \rangle^* \alpha \leq (gf \times Z)^* \gamma$, which is the hypothesis, so $(g, G) \colon (V, Y, \beta_*) \to (W, Z, \gamma)$ is a morphism.

For cartesianness, take $(e, E) \colon (T, S) \to (U, X)$ and suppose the composite (fe, K), with $K(t, y) = E(t, F(et, y))$, is a morphism from (T, S, δ) to (V, Y, β). The square with sides $e \times Y \colon T \times Y \to U \times Y$, $\langle \pi_1, F(e \times Y) \rangle \colon T \times Y \to T \times X$, $\langle \pi_1, F \rangle \colon U \times Y \to U \times X$ and $e \times X \colon T \times X \to U \times X$ is a pullback: a pair (t, x) and (u, y) with $(et, x) = (u, F(u, y))$ is determined by (t, y). In a Heyting category $\forall$ also satisfies Beck–Chevalley along pullbacks, so

$$(e \times X)^* \alpha^* = \forall_{\langle \pi_1, F(e \times Y) \rangle} (e \times Y)^* (f \times Y)^* \beta = \forall_{\langle \pi_1, F(e \times Y) \rangle} (fe \times Y)^* \beta.$$

Hence $\langle \pi_1, E \rangle^* \delta \leq (e \times X)^* \alpha^*$ exactly when $\langle \pi_1, F(e \times Y) \rangle^* \langle \pi_1, E \rangle^* \delta \leq (fe \times Y)^* \beta$. Since $\langle \pi_1, E \rangle \circ \langle \pi_1, F(e \times Y) \rangle = \langle \pi_1, K \rangle$, this is the hypothesis, so $(e, E) \colon (T, S, \delta) \to (U, X, \alpha^*)$ is a morphism. □

Theorem 5.3. *Let $(\mathcal{E}, \mathcal{M})$ be an orthogonal factorization system on $\mathcal{C}$.*

(I) If $\mathcal{C}$ is regular, $\mathrm{Dial}(\mathcal{C})$ has the factorization system whose left class consists of the q-cocartesian morphisms (f, F) with $f \in \mathcal{E}$, and whose right class consists of all morphisms with $f \in \mathcal{M}$ and F fibrewise invertible.

(II) If $\mathcal{C}$ is a Heyting category, $\mathrm{Dial}(\mathcal{C})$ *has the factorization system whose left class consists of all morphisms with $f \in \mathcal{E}$, and whose right class consists of the q-cartesian morphisms with $f \in \mathcal{M}$ and F fibrewise invertible.*

Proof. Combine Lemmas 5.1, 5.2 and 3.2. □

The counterexample side cannot be factored in the same way. Factoring each $F(u,-)\colon Y \to X$ through its image gives a set Z_u depending on u, and the middle object would need a family (Z_w) of counterexample sets indexed by witnesses, which is not a dialectica object. Factorizations of the counterexample side therefore need dialectica objects with counterexamples depending on witnesses, and the epimorphisms of $\mathrm{Dial}(\mathbf{Sets})$ already show this dependence.

Proposition 5.4. *A morphism $(f,F)\colon (U,X,\alpha) \to (V,Y,\beta)$ of* $\mathrm{Dial}(\mathbf{Sets})$ *is*

1. *a monomorphism if and only if f is injective and every $F(u,-)\colon Y \to X$ is surjective;*
2. *an epimorphism if and only if f is surjective and the map $\Phi\colon Y^V \to X^U$ given by $\Phi(H)(u) = F(u, H(fu))$ is injective.*

Proof. For (1), suppose f is injective and every $F(u,-)$ is surjective, and let $(g_1,G_1),(g_2,G_2)\colon (T,S,\gamma) \to (U,X,\alpha)$ have equal composites with (f,F). Then $fg_1 = fg_2$, so $g_1 = g_2 = g$, and $G_1(t,F(gt,y)) = G_2(t,F(gt,y))$ for all t and y; surjectivity of $F(gt,-)$ gives $G_1 = G_2$. Conversely, maps out of $(1,1,\emptyset)$ satisfy the morphism condition vacuously and pick out arbitrary witnesses, which forces f to be injective. For a fixed u, the maps $(u,G_1),(u,G_2)\colon (1,\Omega,\emptyset) \to (U,X,\alpha)$ with $G_1(*,x) = \top$ and $G_2(*,x)$ the truth value of $\exists y\, F(u,y) = x$ are morphisms, and their composites with (f,F) agree because G_1 and G_2 agree on the image of $F(u,-)$. Hence $G_1 = G_2$, and $F(u,-)$ is surjective.

For (2), the composite of (f,F) with $(h,H)\colon (V,Y,\beta) \to (W,Z,\delta)$ is $\big(hf,\ (u,z) \mapsto F(u,H(fu,z))\big)$. If f is surjective and Φ is injective, equal composites give $h_1 = h_2$, and for each z the functions $H_1(-,z)$ and $H_2(-,z)$ have the same image under Φ, so they are equal. Conversely, maps into $(\Omega,\emptyset,\emptyset)$, whose condition is vacuous, force f to be surjective, using the maps $v \mapsto \top$ and $v \mapsto (\exists u\, f(u) = v)$. Maps into $(1,1,\top)$, whose condition always holds, are the pairs $(!,H)$ with $H\colon V \to Y$, and their composites with (f,F) are the pairs $(!,\Phi(H))$, so an epimorphism has Φ injective. □

Classically the condition on Φ is a condition on the fibres of f. Say that the family $\{F(u,-) : f(u) = v\}$ is jointly injective when, for all $y, y' \in Y$, if $F(u,y) = F(u,y')$ for every u with $f(u) = v$, then $y = y'$. If this holds for every v, then Φ is injective: when $\Phi(H) = \Phi(H')$, the elements $H(v)$ and $H'(v)$ agree under every $F(u,-)$ with $f(u) = v$, so $H(v) = H'(v)$. For the converse, given v and y, y' that agree on the fibre over v, take H constantly y and H' equal to y' at v and to y elsewhere; then $\Phi(H) = \Phi(H')$, so $y = y'$. This argument defines H' by cases on whether $v' = v$, so it uses decidable equality on V, and the statement with Φ is the one proved without it.

In particular the left classes of the systems on $\mathrm{Dial}(\mathbf{Sets})$ are not classes of epimorphisms. With $U = V = X = 1$ and $Y = 2$, the map (id,F) with F constant, and with the cocartesian relation β_* on its codomain, lies in the left class of system (I) for the epi–mono system on **Sets**, but is not an epimorphism, since the corresponding map $\Phi\colon 2 \to 1$ is not injective. As with the factorizations, the witness side behaves better than the counterexample side: monomorphisms are described fibre by fibre in both components, while for epimorphisms the condition on counterexamples is fibrewise only when equality on witnesses is decidable.

6 The Girard version inside the Gödel version

The Girard construction maps into the original one, so the factorization systems of the two previous sections can be compared along that map. The functor $J\colon \mathrm{Gir}(\mathcal{C}) \to \mathrm{Dial}(\mathcal{C})$ sends (f, F) to $(f, F \circ \pi_2)$, and the morphism conditions agree because $\langle \pi_1, F\pi_2 \rangle = U \times F$. It is faithful on morphisms out of objects with inhabited witnesses, but not in general: over **Sets** there is a morphism $(\emptyset, X, \emptyset) \to (V, Y, \beta)$ of $\mathrm{Gir}(\mathbf{Sets})$ for each function $Y \to X$, and exactly one in $\mathrm{Dial}(\mathbf{Sets})$.

Proposition 6.1. *Over* **Sets***, with the same system* $(\mathcal{E}, \mathcal{M})$ *on both sides,* J *sends the left class of system (I) on* Gir *into the left class of system (I) on* Dial*. It preserves the system (I) factorization of* (f, F) *when* $F \in \mathcal{M}$*, and, when the witnesses* U *are inhabited, not in general otherwise.*

Proof. The cocartesian relations agree, since $\exists_{f\times Y}\langle \pi_1, F\pi_2\rangle^*\alpha = \exists_{f\times Y}(U \times F)^*\alpha$, which gives the first claim. If $F \in \mathcal{M}$, the Gir factorization of F is $F \circ \mathrm{id}$, the two middle objects are both (W, Y, γ_*), and the factorizations agree. If instead $F = m'e'$ with e' not an isomorphism and U inhabited, then the middle object W is inhabited too, and $J(m, e') = (m, e'\pi_2)$ is not fibrewise invertible, so it is not in the right class on Dial. □

7 Conclusions

A factorization system on the base lifts to both dialectica constructions in two canonical ways, according to whether the middle object of a factorization carries the image of the relation or the largest relation compatible with the second half. For the Girard construction both witnesses and counterexamples factor, and over a Heyting category the two lifts are system (I), whose left maps are cocartesian, and system (II), whose right maps are cartesian, with (I) below (II) and equal to it only when every fibre is trivial. Over sets, starting from epimorphisms and monomorphisms, they are (regular epimorphisms, monomorphisms) and (epimorphisms, regular monomorphisms). Since $\mathrm{Gir}(\mathbf{Sets})$ is not balanced it has no (epimorphism, monomorphism) system, and its regular epimorphisms need not split even though **Sets** satisfies choice. All of this is proved constructively.

For the original construction the witness side factors as before, with right maps whose counterexample components are fibrewise invertible, and the same two lifts of relations give systems (I) and (II) on $\mathrm{Dial}(\mathcal{C})$. The counterexample side does not factor, because the image of each $F(u, -)$ depends on u. The same dependence appears in the epimorphisms of $\mathrm{Dial}(\mathbf{Sets})$, whose condition on counterexamples is the injectivity of $\Phi\colon Y^V \to X^U$ and becomes a condition on fibres only when equality on witnesses is decidable. Both phenomena call for dialectica objects whose counterexamples vary with witnesses, where the counterexample side could be factored as well.

Three things were wanted and not reached here. Example 4.3 shows that, relative to system (I), both "every epimorphism splits" and "every epimorphism has a monomorphism going back" fail in $\mathrm{Gir}(\mathbf{Sets})$ for reasons that have nothing to do with choice in **Sets**, while the set-theoretic statements of [3] live in subcategories such as PV_X; which choice and partition principles become visible in such subcategories, relative to the systems described here, is the question this note was written for. How the systems interact with the adjunction between $\mathrm{Gir}(\mathcal{C})$ and $\mathrm{Dial}(\mathcal{C})$ in [5] has not been worked out, and neither has the extension to counterexamples depending on witnesses.

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